**Chemistry 3A – Spring 2026** **Version A**

Sections 19320-19321

Midterm Examination #3

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt.
TEAR OFF LAST PAGE and use it as scratch for writing calculations.

1. Convert a pressure (*P*) of **760 torr** to millimeters of mercury (**mmHg**)
   a) 273.15 mmHg b) 1 mmHg c) 760 atm d) 760 mmHg

2. Which gas law defines the mathematical relationship $V_1/T_1 = V_2/T_2$?
   a) Boyle's Law b) Charles' Law c) Gay-Lussac's Law d) Ideal Gas Law

3. Which of these statements is NOT part of principles & assumptions of Kinetic Molecular Theory of Gases?

a) a gas is composed of tiny particles in constant motion
b) gas particles collide elastically: not losing energy in collisions with each other and container walls
c) Gas particles are separated by distances small relative to their size
d) the average speed is dependent on temperature

4. A gas has an initial pressure $P_1$=1.00 atm and volume $V_1$ = 2.00 L. The pressure is reduced to $P_2$ = 0.800 atm. What is new volume $V_2$ in milliliters (mL) if all else (temperature, moles) is constant?
   a) 2.50 mL b) 1,600 mL c) 2,500 mL d) 4,000 mL

5. An ideal gas of *n* =2.00 mol at a temperature *T* = 25°C has a pressure of P = 1.50 atm. What is the volume in liters (L)?
   a) 2.74 L b) 24.5 L **c) 32.6 L** d) 48.9 L

6. An aqueous solution of calcium chloride ($CaCl_2$) is prepared. Which of these is a ***false*** statement about this solution?
   a) Calcium chloride ($CaCl_2$) is the *solute* and water ($H_2O$) is the *solvent*
   b) In nonsaturating aqueous solutions, the salt dissolves fully to form $Ca^{2+}$ and $Cl^-$ ions
   c) Water ($H_2O$) is better than hexane ($C_6H_6$) in choosing to dissolve calcium chloride
   **d) Calcium chloride forms a precipitate when dissolved in water**

7. Which of the statements below is a ***false*** statement regarding *electrolytes*?
   a) The ionic solute dissolving will make a very good conductor of electric current
   b) The ionic compounds that dissolve in solution do so (almost) entirely
   c) A salt is a good electrolyte if it dissolves to form ions in solution
   d) The best electrolytes are molecules like carbon dioxide ($CO_2$) and atmospheric nitrogen ($N_2$)

8. Which of these reactions of silver nitrate ($AgNO_3$) and sodium bromide (NaBr) shows the **complete ionic equation**?
   a) $AgNO_3$ (aq) + NaBr (aq) → AgBr (s) + $NaNO_3$ (aq)
   **b) $Ag^+$ (*aq*) + $NO_3^-$ (*aq*) + $Na^+$ (aq) + $Br^-$ (*aq*) → AgBr (*s*) + $Na^+$ (*aq*) + $NO_3^-$ (*aq*)**
   c) $Ag^+$ (*aq*) + $Br^-$ (aq) → AgBr (*s*)
   d) NONE of the above show a **complete ionic equation**

9. A dilution of a $C_1$ = 2.00 *M* KCl stock solution was used to make a $V_2$ = 200.0 mL of a $C_2$ = 0.500 M KCl solution. What stock volume aliquot $V_1$ was used?
   a) 0.005 mL b) 50 mL c) 200 mL d) 800 mL

10. Which of the statements below refer to the meaning of "like dissolves like"?
  a) It refers to combustion reactions with oxygen in the atmosphere
  b) It means that the compound calcium fluoride ($CaF_2$) will dissolve in a nonpolar solvent like octane ($C_8H_{18}$)
  c) It means polar solutes will dissolve in polar solvents
  d) It means that gases whose volume is reduced by half will see their pressures reduced by half

11. Which of the following is true about pressure as related to gases?
  a) It is a factor in Boyle's Law
  b) It is understood in physics as a force per unit area (P=force/area)
  c) A measured pressure can be expressed in units of millimeters of mercury (mmHg)
  d) all of the above are true

12. Which of these solutes will have a greater effect on elevating the boiling point temperature of a solvent given they completely dissolve?
  a) KCl  b) $CaCl_2$  c) ethanol ($CH_3CH_2OH$)  d) glucose ($C_6H_{12}O_6$)

13. Which of the following is FALSE about written chemical reactions?
  a) Conservation of mass and charge is required in the balancing of chemical reactions
  b) The chemical compounds to the left of the arrow (yield symbol) are called "products"
  c) A polyatomic molecule should be shown in parentheses if present more than once in a compound
  d) The number and type of atoms on both sides of the reaction arrow (⟶) MUST be equal

14. In written chemical reaction equations, what do uses of (*s*) and (*aq*) indicate?
  a) the reaction is not stoichiometric (it has yet to be balanced)
  b) they are references to the state or "phase" of the compound/matter
  c) they show whether a reaction is endothermic or exothermic
  d) they are only used in half-reactions (oxidation-reduction reactions)

15. Precipitation reaction $AgNO_3$ (*aq*) + NaCl (*aq*) ⟶ AgCl (*s*) + $NaNO_3$ (*aq*) is shown. What is true about balancing this reaction?
  a) Balancing requires changing the chemical reaction to add coefficient of "2" to $AgNO_3$
  b) The reaction equation should only ever be balanced by adjusting the whole-number coefficients of the compounds
  c) Removing AgCl from the reactants would make the equation balanced
  d) Balancing requires changing AgCl from solid (*s*) to aqueous (*aq*) to match other products

16. Which reaction below shows acid-base neutralization type reaction
  a) Mg (*s*) + $Cu(NO_3)_2$ (*aq*) → $Mg(NO_3)_2$ (aq) + Cu (*s*)
  b) HCl (aq) + NaOH (aq) → NaCl (aq) + $H_2O$ (*l*)
  c) $F_2$ (g) + $MgCl_2$ (s) → $Cl_2$ (g) + $MgF_2$ (s)
  d) $CaCO_3$ (s) ⟶ CaO (*s*) + $CO_2$ (*g*)

17. 1.00 mol of Li chloride (LiCl) [molar mass = 42.394 g/mol] is dissolved in final 1.00 kg water. What is the **molality** of the LiCl solution?
  a) 0.0236 *m*  b) 1.00 *m*  c) 42.4 *m*  d) 1,000 *m*

18. An aqueous solution is prepared with 100.0 g of sodium bromide (NaBr) [molar mass = 102.89 g/mol] to final value of 500.0 mL with water. What is the concentration of the NaBr solution?
  a) 0.2000 M  b) 0.4860 M  c) 1.944 M  d) 2.058 M

19. Which reaction below shows a **metal replacement** (single replacement) reaction type?
  a) $Cl_2$ (*g*) + 2 NaBr (*aq*) ⟶ 2 NaCl (*aq*) + $Br_2$ (*l*)
  b) Mg (*s*) + $Cu(NO_3)_2$ (*aq*) ⟶ $Mg(NO_3)_2$ (*aq*) + Cu (*s*)
  c) $BaCl_2$ (*aq*) + $Na_2SO_4$ (aq) ⟶ $BaSO_4$ (*s*) + NaCl (*aq*)
  d) $H_2O$ (*l*) ⟶ $H_2$ (*g*) + 2 $O_2$ (*g*)

20. This is written on the whiteboard: "$I_2 + 2\ e^- \longrightarrow 2\ I^-$". What is true about what is written?
  a) This shows an oxidation half-reaction because $I_2$ is gaining electrons
  b) This reaction is not possible because chlorine (Cl) is missing
  c) Shows a reduction half-reaction in which iodine (I) had oxidation number 0 and now it is -1
  d) The number of protons should be shown on the right side of the half-reaction

21. Which statement is FALSE with respect to **colligative properties** of a solution?
  a) These properties depend only on the **numbe**r of solute particles dissolved in the solvent.
  b) **Molality** (*m*) is the required unit of concentration for calculating freezing and boiling point changes.
  c) A solution's change in boiling point and freezing point are examples of colligative properties
  d) To understand the effect of the solute, you must know the **oxidation numbers** of the atoms in the compound.

22. Find the TRUE statement concerning the **oxidation number** of molecule nitrogen dioxide ($NO_2$)?
  a) The oxidation number of oxygen (O) atoms is 0 (zero)
  b) The sum of all oxidation numbers for this $NO_2$ is 7
  c) The oxidation number of the nitrogen (N) atom is **+4**
  d) Hydrogen is missing so the oxidation number for any N or O cannot be determined

23. Sodium (Na) metal will react explosively with cold water in a vigorous redox reaction. Which of the reactions below is correctly balanced and shows what occurs?
  a) 2 Na (*s*) + 2 $H_2O$ (*l*) ⟶ 2 $O_2$ (*g*) + 3 NaOH (*aq*)
  b) Na (*s*) + 3 $H_2O$ (*l*) ⟶ $H_2$ (*g*) + NaOH (*aq*)
  c) 2 Na (*s*) + 2 $H_2O$ (*l*) ⟶ $H_2$ (*g*) + 2 NaOH (*aq*)
  d) Na (*s*) + $H_2O$ (*l*) ⟶ $O_2$ (*g*) + NaOH (*aq*)

24. What happens to the solubility of most gases in water as temperature decreases?
  a) It increases
  b) It decreases
  c) It remains constant
  d) It depends on the volume only

25. When heated, potassium chlorate will produce potassium chloride and oxygen gas. The reaction is 2 $KClO_3$ (s) → 2 KCl (s) + 3 $O_2$ (g). What type of reaction is this?
  a) combustion
  b) acid-base neutralization
  c) decomposition
  d) precipitation

TRUE or FALSE questions. (a) for true, (b) for false

26. A decomposition reaction involves one compound forming two or more product compounds
    a) true  b) false

27. The oxidation number of oxygen atom in most cases is 2– based on its electronegativity.
    a) true  b) false

28. Ethanol ($CH_3CH_2OH$) is a good example of a strong electrolyte.
    a) true  b) false

29. The molarity of a solution is expressed in units of an italicized capital M (*M*).
    a) true  b) false

30. A precipitate is an insoluble compound (a substance forming a solid from solution).
    a) true  b) false

31. The equation $PV = nRT$ is also known as Avogadro's Law of Gases.
    a) true  b) false

32. In a chemical reaction equation, compounds to ***left*** of the ***reaction arrow*** are called **reactants**.
    a) true  b) false

33. Ionic compounds prefer to dissolve in polar solvents like water, not in nonpolar solvents.
    a) true  b) false

34. When an element gains electrons in a reaction, this is called a **reduction**.
    a) true  b) false

35. At Standard Temperature and Pressure (STP), the temperature is 0 K and the pressure is 1 atm.
    a) true  b) false